Veera Naga Koushik Chaladi

Role: Senior Associate Software Engineer

Contact Number: +91 9110753275, **Email:** vnkoushik.ch@gmail.com

Results-driven AI/ML Engineer with 4.9 years of experience in AI-driven automation, machine learning, and backend API development. Adept at designing **multi-agent systems, retrieval-augmented generation (RAG) models, and optimizing automation pipelines** to enhance efficiency **Proficient in Python, FastAPI, OpenAI APIs, Machine learning models and** with goods hands experience on **AWS, Devops setup** with a proven track record of delivering AI-powered automation solutions.

Professional Experience:

Amgen

Senior Associate Software Engineer

Project: MAP AI Assistant

- Developed a **chat-based agentic AI system** to automate employee review generation, including employee self-evaluations, goals, behaviors, and overall performance reviews. Based on competencies for different roles
- Implemented a **RAG-based model** to identify recognitions received by employees and search through files uploaded by employees in chat.
- Reduced **AWS infrastructure costs by 29.4%** through optimization and resource tuning strategies.
- **Optimized DynamoDB** usage, achieving an additional **10%** cost reduction.
- Took ownership of **technical leadership activities**, guiding the team on architecture, implementation, and technical decision-making.
- Was instrumental in setting up **devops pipeline** from scratch and handled devops pipeline issues.
- Worked on **Redis caching mechanism** to store the chat data in memory instead of hitting DB.

Project: AI Agent for Scientific Results

- Developed an **AI agent to interpret scientific results** by extracting key findings from analysis outputs in the input deck.
- Connected **internal scientific results with relevant literature** and public evidence to support interpretation and validation.
- **Optimized performance of agent by 30%** by implementing **Planner and Executor** agentic AI pattern with parallel execution of independent acyclic DAGs.
- **Assessed known versus novel findings**, generated mechanism-linked interpretations, and produced traceable insight summaries.
- Used **PubMed MCPs and vector database search** to identify concrete evidence from scientific literature.
- Was responsible for debugging and fixing devops pipeline issues.

Project: AI Agent for Scientific

Pegasystems

Applications Engineer

Project: Pega Autopilot *Mar 2023 to June 2025*

- Developed **Pega AutoPilot**, an AI-driven automation system leveraging **AutoGen, Multi-Agent**

Communication, RAG models, and FastAPI to streamline pega case management

- **Multi-Agent System for Intelligent Task Execution** – Designed a multi-agent communication framework, ensuring efficient collaboration between AI assistant and agents.
- **Knowledge Retrieval (RAG Model)** – Implemented retrieval-augmented generation (RAG) to fetch contextual data when required data is unavailable, enabling iterative AI-system interactions.
- Developed **FastAPI** endpoints for efficient API communication and automation triggering.
- Developed modular plugin-based architecture, improving automation success rates by 20% and reducing open ai costs
- **Optimized API token** usage, cutting costs by 35% through streamlined function execution and reducing unnecessary model calls.

Associate Applications Engineer

Project: ML Model execution service *Aug 2022 to Mar 2023*

- **Automated Machine Learning Model Creation** – Enabled users to upload data and models via Pega Infinity UI, triggering automated ML workflows in the backend.
- **Ensemble Model Implementation** – Developed and optimized ensemble models, improving predictive accuracy and generalization.
- **Machine Learning Pipeline Development** – Designed a scalable ML pipeline, ensuring seamless preprocessing, model training, and evaluation.
- **Hugging Face Model Integration** – Integrated Hugging Face transformers, enabling advanced NLP model execution within Pega AI.
- **Sklearn-Based Model Creation** – Utilized Scikit-Learn for classical machine learning model development and evaluation.

Project: Smart claims engine *Aug 2021 to Jul 2022*

- **Pega Case Management & Background Processing** – Expertise in case types, service REST, file listeners, collection rules, report definitions, and product rules for streamlined workflow automation
- Advanced Reporting & Search Optimization – Developed virtual report definitions for Elastic Search, Pega Search, and Reporting Service, implemented near real-time indexer using Indexer API, and optimized advanced paging functionality.
- **Declarative & Data Processing** – Hands-on experience with declarative rules, activities, data transforms, joins, associations, sub-reports, and XML parsing to enhance automation and data integrity.
- **Pega AI & Integration** – Integrated Pega Gen AI functionality, leveraged Pega Explore Data, and executed Service REST API implementations for seamless data processing and retrieval.
- **POC Development** – Contributed to design discussions, EPICs POCs, and enhanced system extensibility

Skills

Programming/Scripting Languages: Python,

Machine Learning: PyTorch, Tensorflow, Scikit-Learn, Ensemble learning

AI & NLP expertise: LLMs, Transfomers, Hugging face, LangChain, Spacy, OpenAI API, nltk

Backend API & databases: FastAPI, SQL

Others: GIT, GenAI, Prompt Engineering, AWS, Docker, Basics of Linux

Soft Skills: Team Player, Resilient, Problem Solving, Inter-Personal Communication.

Education

Velagapudi Ramakrishna Siddhartha Engineering College (VRSEC)

Bachelor of Technology – Information Technology; (GPA – 8.45) *May 2017 to Jun 2021*

Certifications

Google tensorflow developer certification	May 2023 – No expiry
Pega certified senior system architect	Mar 2021 – No expiry

Awards and Recognitions

- 2-time spot award winner.
- Winner of pega **4x4 award** which is awarded globally in each quarter
- Published Journals in Indexed, Scopus, and Free journal for springer.
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